



Figure 3: How AWS IoT and Salesforce IoT work together (Image source: [googleimages.com](https://www.googleimages.com))

are similar according to some pre-designated criteria. But all the observations that are drawn from different clusters are dissimilar. Different clustering techniques use different assumptions on the structure of data. This is often defined by using some similarity metrics and is then evaluated. It is an unsupervised learning method.

Bayesian networks: This is a kind of probabilistic graphical model that represents a set of random variables with all their conditional independencies by making use of a directed acyclic graph. For instance, it can represent the probabilistic relationships present between various diseases and their symptoms. If we know the symptoms, then this network can easily calculate the probability of the presence of various diseases.

Decision tree learning: Decision tree learning is a type of ML technique that uses a decision tree as its predictive model. This model maps the observations about an item to different conclusions about the target value of the item.

Association rule learning: This is actually a method used for discovering different interesting relations between all the variables present in large databases.

K-nearest neighbours: This is a supervised learning algorithm which classifies a data sample (also called a query point) on the basis of the labels (also called output values) of the nearby data samples. Basically, this algorithm classifies the K kinds of clusters such that the distance inside is minimal. This is a type of general classification algorithm.

Application of ML in IoT

In the energy sector: Arduino MEGA is used to reduce the energy cost for a coffee machine. This is one of the simpler implementations of ML with IoT. But these techniques are also being implemented in lights and air conditioners connected to the IoT.

In routing traffic: The combination of different sensors and ML algorithms is widely implemented for routing traffic. One such implementation is in a system using the LarKC platform, shown in Figure 4. The data from

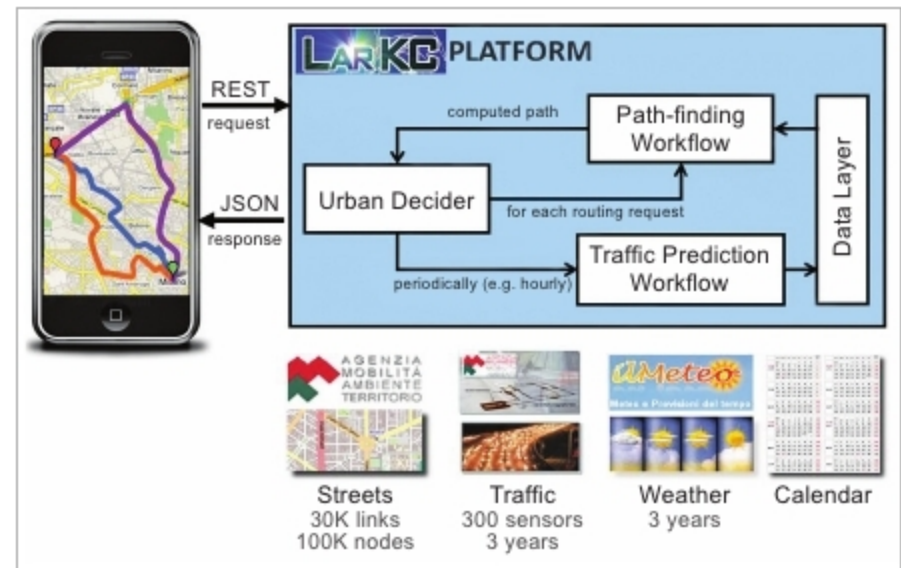


Figure 4: ML and IoT used in routing traffic (Image source: [googleimages.com](https://www.googleimages.com))

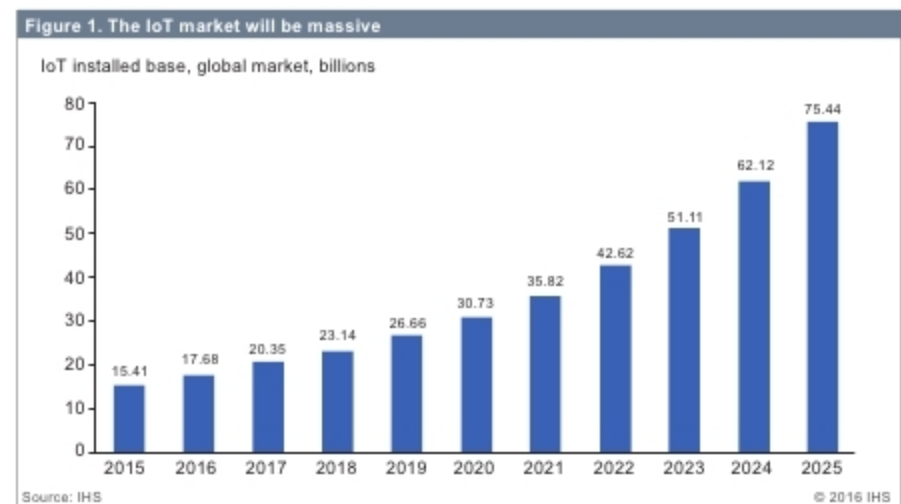


Figure 5: IoT market statistics for the future (Image source: [googleimages.com](https://www.googleimages.com))

the traffic and weather can suggest several routes to the same destination.

In the home: IoT applications are extremely suitable for the home and are being widely used in home automation. They are implemented in apartments for light and humidity control as well as with temperature sensors. IoT is also applied in heart rate sensors.

In industry: There are several companies and organisations that use IoT and ML algorithms for health care, traffic management, etc. In manufacturing industries, considerable human resources can be saved with the help of an IoT system that uses cameras and controllers. **END** 🐧

References

- [1] <http://www.wikipedia.org/>
- [2] <https://www.sas.com/>
- [3] <https://dzone.com/>
- [4] <https://www.wired.com/>

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